



SRI SHAKTHI

INSTITUTE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution

Coimbatore - 62.



NATIONAL BOARD
OF ACCREDITATION
For CSE, MECH, EEE, IT



SMART HELPDESK FOR IT SERVICES

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AGENDA

- OBJECTIVE
- ABSTRACT
- INTRODUCTION
- LITERATURE SURVEY
- EXISTING SYSTEM AND ITS DRAWBACKS
- PROPOSED SYSTEM
- TECH STACK
- REFERENCE
- CONCLUSION

OBJECTIVE

- To automate IT service request handling using AI for faster ticket classification and it's resolution.
- To provide a centralized platform with chatbot support for easy ticket creation and tracking.
- To improve service efficiency and user satisfaction through real-time alerts and smart analytics.

ABSTRACT

- ❑ The Smart Helpdesk Ticketing System is an AI-based platform that automates IT service request handling through web, mobile, and chatbot interfaces.
- ❑ The system improves response time, service efficiency, and user satisfaction with real-time updates and performance analytics.
- ❑ The system is scalable and user-friendly. Overall, it ensures faster, smarter, and more efficient IT service management.

INTRODUCTION

- Modern organizations face challenges in managing large volumes of IT service requests efficiently.
- A helpdesk system acts as a centralized support platform that helps organizations manage IT issues, service requests, and user communication efficiently.
- An AI-based smart helpdesk system can automate support processes and improve service quality.

LITERATURE SURVEY

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2023	Yashraj Chanchad, Suraj Kanade, Ranjana Singh	Web-Based Helpdesk Ticketing System	Web Development, Database Systems	No AI automation; manual classification

LITERATURE SURVEY

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2024	Dr. S. Parvathi, P. Ajith, J. Daniel Tharmaraj, T. R. Dharaneeshl	Multilingual Chatbot Ticketing System	NLP, AI Chatbot, Analytics Dashboard	High development complexity

LITERATURE SURVEY

YEAR	AUTHOR	TITLE	TECHNOLOGY USED	LIMITATION
2025	S. Manonmani, V. Deepa, N. Dhivyabharathi, P. Harshini, M. Akalya	AI Chatbot-Based Ticketing System	NLP, Machine Learning, Chatbot Systems	Limited real-time scalability; depends on training data

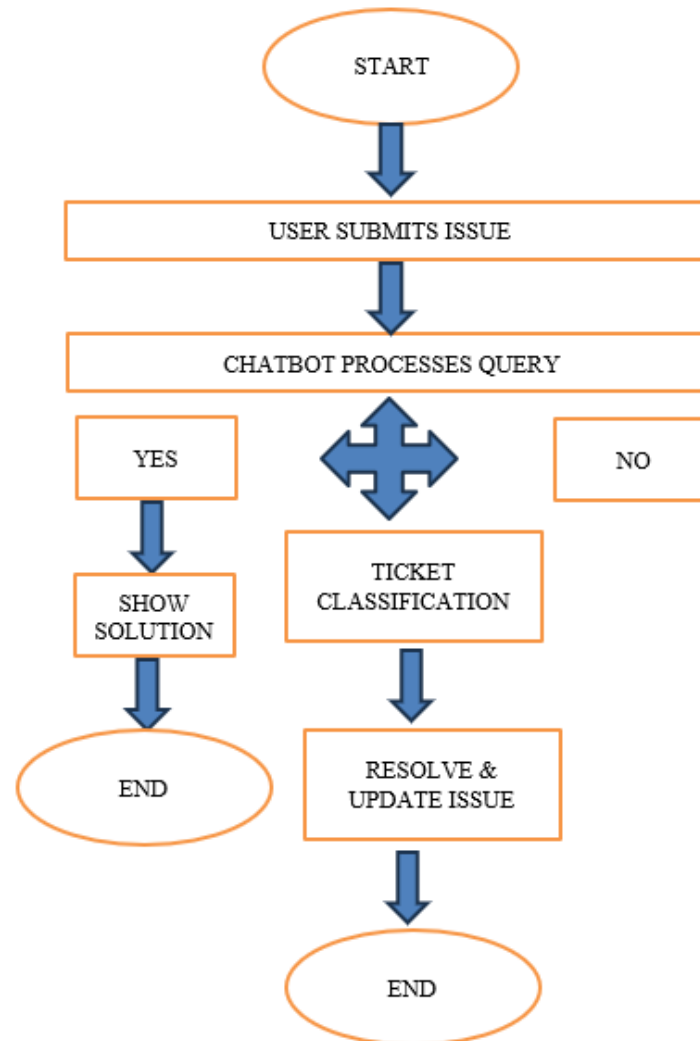
EXISTING SYSTEM AND ITS DRAWBACKS

- IT service requests are mostly handled manually through emails, calls or basic ticketing tools.
- Support staff manually categorize and assign tickets to respective departments.
- Delays in issue resolution due to manual processing.
- Higher workload on IT staff and chances of human errors.

PROPOSED SYSTEM

- An AI-based smart helpdesk platform that automates ticket creation, classification and assignment.
- Integration of an intelligent chatbot to assist users and provide instant solutions.
- A centralized dashboard for real-time ticket tracking and performance monitoring.
- Smart knowledge base that learns from past issues and resolutions.

FLOW CHART



TECH STACK



Frontend:

- HTML, CSS, JavaScript – for user interface design
- React.js – for dynamic and responsive UI
- Tailwind CSS – for clean styling



Backend:

- Node.js – server-side logic
- Express.js – API handling
- MongoDB – database for storing tickets and user data



AI:

- Gemini Flash 2.0 API – chatbot & ML services
- NLP libraries – for ticket classification

HELPDESK & AI INTEGRATION

- **Automated Escalation System** – Escalates unresolved tickets based on SLA time limits.
- **Role-Based Access Control (RBAC)** – Secure access for Admin, Support Staff, and Users.
- **AI-Based Ticket Classification using NLP** – Automatic issue understanding and smart categorization.
- **Chatbot Integration with Context Awareness** – Intelligent bot that understands user queries and provides instant solutions

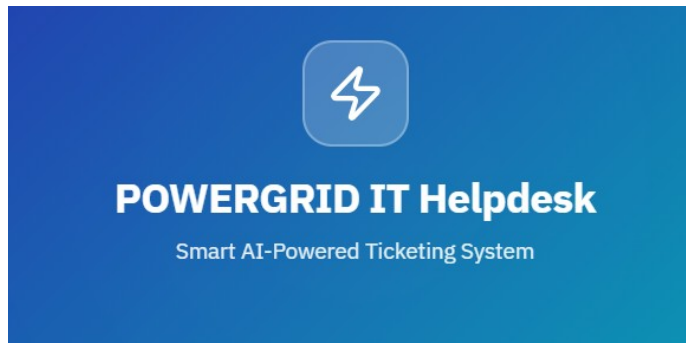
REFERENCE

- [1] Y. Chanchad, S. Kanade, and R. Singh, "Helpdesk Ticketing System," *International Journal of Innovative Research in Technology (IJIRT)*, vol. 9, no. 3, pp. 120–125, 2023.
- [2] S. Parvathi, P. Ajith, J. D. Tharmaraj, and T. R. Dharaneeshl, "Online Chatbot-Based Ticketing System," *International Journal of Scientific Research and Engineering Trends (IJSRET)*, vol. 10, no. 6, pp. 508–512, 2024.
- [3] S. Manonmani, V. Deepa, N. Dhivyabharathi, P. Harshini, and M. Akalya, "AI Chatbot-Based Ticketing System," *International Journal of Scientific Research in Engineering and Management (IJSREM)*, vol. 9, no. 2, pp. 1–6, 2025.
- [4] A. M. Lakshmi, C. S. Deepika, A. Arshad, and J. K. B, "Online Chatbot-Based Ticketing System," *IRE Journals*, vol. 8, no. 5, pp. 45–50, 2025.
- [5] R. Kumar and S. Patel, "AI-Based Customer Support Automation," *International Journal of Scientific Engineering and Technology (IJSET)*, vol. 12, no. 5, pp. 822–828, 2024.

CONCLUSION

- The proposed system improves IT support by using **AI-based automation and smart ticket management**.
- The integration of **chatbot, NLP, and machine learning** enhances user experience and efficiency.
- Overall, the system provides a **scalable, fast, and intelligent helpdesk solution** for modern organizations.

SIGN UP AND LOGIN PAGE



Login

Sign Up

Employee ID / Email

helpdesk@gmail.com

Password

.....

[Forgot Password?](#)

Login to Dashboard

Full Name

HELPDESK

Employee ID

24CS299

Email Address

helpdesk@gmail.com

Password

.....

Password strength: Strong

Confirm Password

.....

Account Role

Employee (TICKETER)

Employee (TICKETER)

Support (Ticket Handler)

Admin

ADMIN DASHBOARD



POWERGRID IT Helpdesk

Smart AI-Powered Ticketing System

A

ADMIN CONTROLS

 Admin Dashboard

 All Tickets

 Employees

 Support Staff

 Configuration

Team Management

 Full Analytics

Welcome Admin Arun!

You have full control over the POWERGRID IT Helpdesk ecosystem. Select a module from the sidebar to manage standard tickets, configuration, team roles, or system analytics.



All Tickets

Manage global employee tickets

HANDLER DASHBOARD



POWERGRID IT Helpdesk

Smart AI-Powered Ticketing System

P

SUPPORT DESK



Handler Dashboard



All Tickets



My Team



Team Analytics

Welcome Poovarasank!

You have full control over the POWERGRID IT Helpdesk ecosystem. Select a module from the sidebar to manage standard tickets, configuration, team roles, or system analytics.



All Tickets

Manage global employee tickets



EMPLOYEE DASHBOARD



POWERGRID IT Helpdesk

Smart AI-Powered Ticketing System

+ Create Ticket

NP

Dashboard

+ Create Ticket

My Tickets

RESOURCES

Knowledge Base

Notifications

Dashboard Overview

Welcome back, **Navaneeth P!** Here's your IT helpdesk summary for today.



~ --%

--

Total Tickets

23 open · 12 in progress



~ --%

--

Resolved Tickets

8 resolved today



~ 5%

--

Avg. Resolution Time

Target: 6 hours



~ --%

--%

SLA Compliance

5 breached this week



TICKET CREATION PAGE



POWERGRID IT Helpdesk

Smart AI-Powered Ticketing System

+ Create Ticket

NP



Dashboard



Create Ticket



My Tickets

RESOURCES



Knowledge Base



Notifications

Create New Ticket

Describe your issue and our AI will help route it to the right team



AI Ticket Creation

Describe your issue in a single sentence, and AI will auto-fill the ticket details.

[AI] Slow Network Performance Reported

Auto-Fill

Issue Title *

slow Network Performance

Category *

Network

Priority

Medium - Should be resolved soon



AI Suggestions

Based on your description, here are some quick solutions:

Try Password Reset

Most access issues can be resolved by resetting your password

Check System Status

Verify if there's a known outage affecting your service

Clear Browser Cache

Many login issues are resolved by clearing browser data



ANALYTICS PAGE



Analytics & Reports

POWERGRID AI Helpdesk Intelligence Dashboard

Export PDF

Dashboard

P

System Analytics

Comprehensive performance metrics for POWERGRID IT Helpdesk

7 Days

30 Days

90 Days

1 Year



+12%

1,247

Total Tickets
This period



+8%

94.8%

SLA Compliance
Target: 95%



-15%

4.2h

Avg. Resolution
Target: 6h



+22%

38.4%

AI Auto-Resolved
479 tickets



-33%

5.2%

SLA Breach Rate
65 breaches



+5%

4.7/5

User Satisfaction
892 ratings



AI-Powered Insights

Automatically generated from ticket patterns this period

Peak Hours Detected

Ticket volume spikes 68% between 10–12 AM on Mondays. Consider adding on-call staff during this window.

Network Issues Rising

Network tickets increased by 35% this month. Building A is the hotspot — infrastructure review recommended.

AI Saving 40% Time

AI chatbot deflected 479 tickets that would otherwise require engineer intervention, saving ~2,000 man-hours.

SLA Risk Detected

Critical tickets on Fridays have a 3× higher SLA breach rate. A dedicated on-call engineer is advised.



Ticket Volume Trend

Real-time data from the last 30 days

Weekly

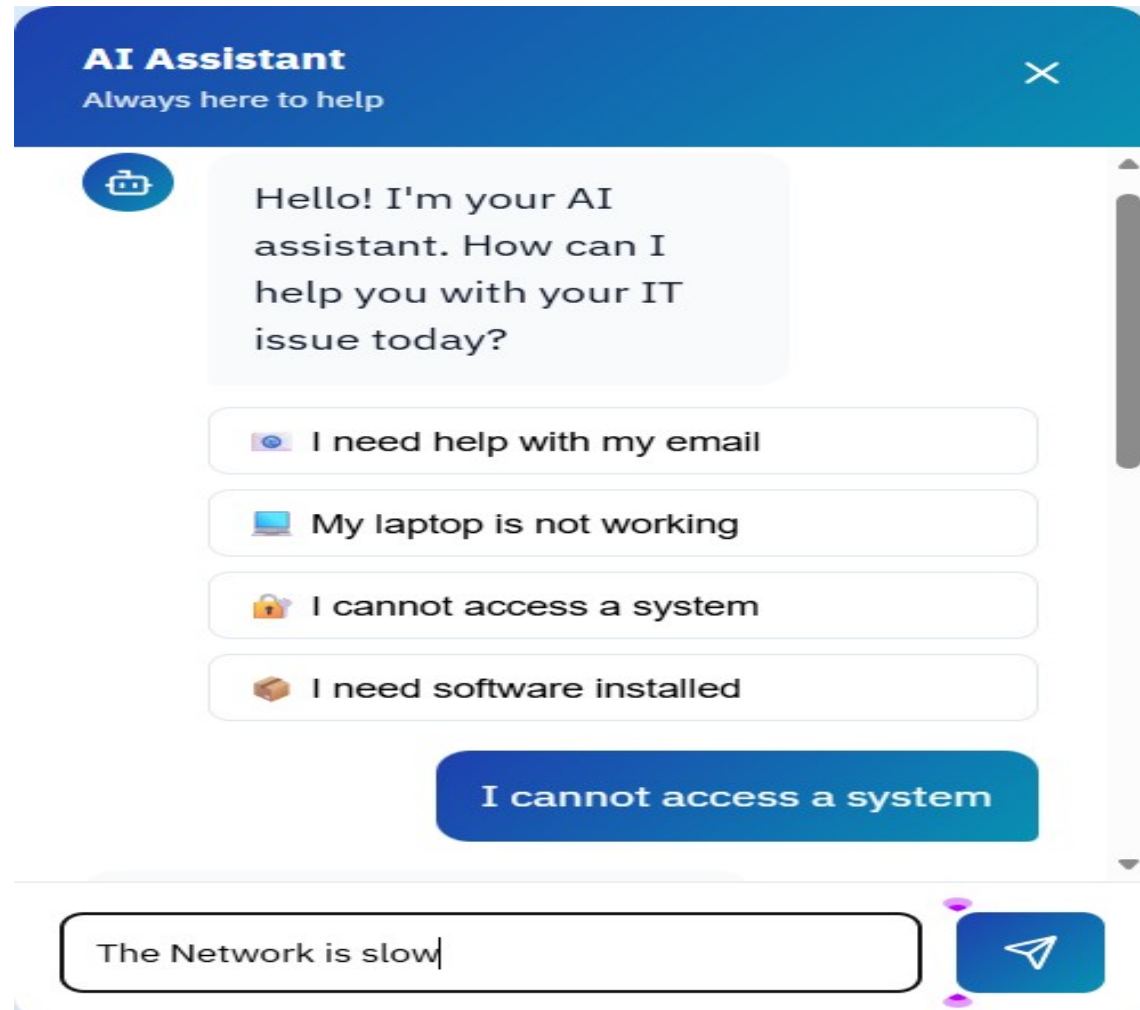
Daily



Tickets by Category

Distribution this period

AI CHATBOT ASSISTANT

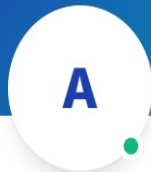


PROFILE



POWERGRID IT

← Dashboard



Arun

arun@gmail.com

Admin

Active

STATUS

24cs188

EMPLOYEE ID

[→ Secure Logout]



Account Settings & Identity

VERIFIED CREDENTIALS

Full Name

Arun

Email Address

arun@gmail.com

CONTACT & COMPANY INFO

Mobile Number *

+919788361763

Department / Batch

Computer Science

Date of Birth

10-06-2006

Nationality

india

Home Address *

5/5-L&T ByPass, Chinniyampalayam , Coimbatore-59

GOVERNMENT IDENTITY

THANK YOU!!!